TIC 119941327

Eclipsing binary candidate dossier - Star Vetter

PERIOD ALIAS

```
BLS EPHEMERIS
  Period       3.260646 d  (ALIASED, see below)
    -> corrected period: 6.521291 d
  Epoch (t0)   1470.0153 BTJD
  Duration     0.1700 d
  Depth        4.227 %  (42,270 ppm)
  SNR          116.8

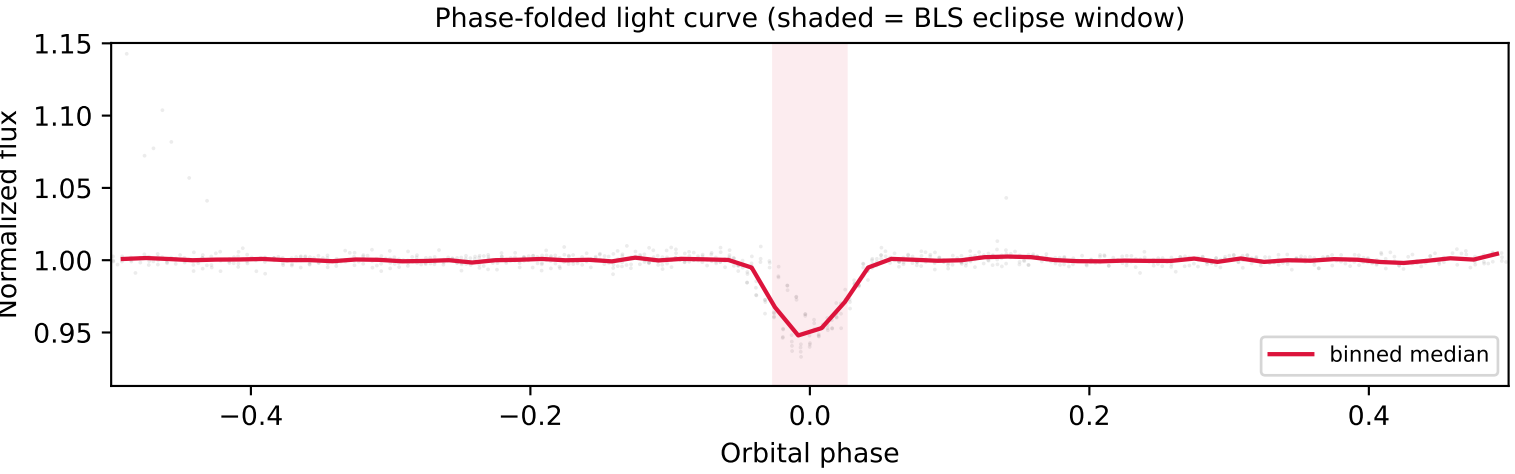
INDEPENDENT CHECKS (f3i gates)
  Eclipses observed     7  (gate >= 3)
  Robust depth ratio    1.04  (gate >= 0.50)
  Robust depth sigma    87.6  (gate >= 7.0)
  Odd/even z-score      3.38  (ALIAS FLAGGED)
  Secondary eclipse     not detected
```

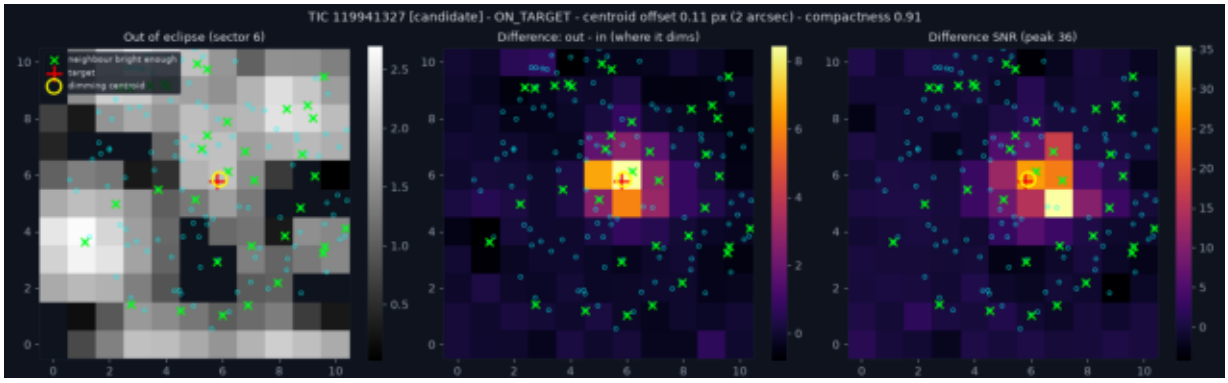
```
CATALOG CROSS-MATCH (novelty)

  VSX                                 no match
  ASAS-SN                             no match
  Gaia eclipsing-binary table         no match
  Gaia variable classification        no match
  TESS EB catalog (Prsa+2022)         no match
  ExoFOP TOI list                     no match

  Verdict: NOVEL
```

```
PIXEL-LEVEL SOURCE CHECK (TESScut difference imaging)
  Sector 6    light-curve dip in this sector: 84.6 sigma    t0 refined by -0.0025 phase
  Difference-image peak SNR   35.51   (gate >= 5.0)      Compactness   0.91   (gate >= 0.50)
  Dimming centroid offset     0.11 px (2 arcsec)   (gate <= 1.0 px, i.e. on the target)
  Nearest bright-enough neighbour in cutout: TIC 119941337
    0.41 px away, dTmag +3.16 - farther from the dimming than the target
```





Caveat (PERIOD ALIAS): Odd/even eclipse-depth z-score is 3.38 (gate > 3.0): alternating eclipses differ significantly in depth, the signature of a true period double what BLS found. Pixel-check confirmation and eclipse timing are unaffected - report the corrected period, 6.521291 d, in any write-up.